Correct Marks : 4 Selectable Option : 0

Question Label : Multiple Select Question

Identify the sets of valid test cases for branch coverage.

Options :

6406531515283. ✖ {day = 0, day = 5}

6406531515284. ✖ {day = -1, day = 5, day = 8}

6406531515285. ✔ {day = 0, day = 5, day = 7}

6406531515286. ✔ {day = 5, day = 6, day = 9}

Question Number : 161 Question Id : 640653455719 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4

Question Label : Multiple Choice Question

Identify the sets of valid test cases for multiple condition coverage.

Options :

6406531515287. ✔ {day = 0, day = 8, day = 5, day = 7}

6406531515288. ✖ {day = -1, day = 8, day = 5, day = 1}

6406531515289. ✖ {day = 0, day = 5, day = 7}

6406531515290. ✖ {day = 5, day = 6, day = 9}

Speech Tech

Section Id :	64065329451
Section Number :	7
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	7
Number of Questions to be attempted :	7
Section Marks :	40

Display Number Panel :	Yes
Group All Questions :	No
Enable Mark as Answered Mark for Review and Clear Response :	Yes
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	64065365964
Question Shuffling Allowed :	No
Is Section Default? :	null

Question Number : 162 Question Id : 640653455738 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DEGREE LEVEL: SPEECH TECHNOLOGY"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?
CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECT TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT ,PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406531515364. ✔ Yes

6406531515365. ✖ No

Sub-Section Number :	2
Sub-Section Id :	64065365965
Question Shuffling Allowed :	No
Is Section Default? :	null

Question Number : 163 Question Id : 640653455739 Question Type : MCQ Is Question

Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 3

Question Label : Multiple Choice Question

1.1 MCQ

1. (1 point) For a given set of data points $X = \{x_1, x_2, ..., x_N\}$, log-likelihood for a GMM with k mixtures can be defined as:

   A. $\sum_{j=1}^{N} \log(\sum_{i=1}^{k} \pi_i N(x_j, \mu_i, \sigma_i^2))$

   B. $\log(\prod_{j=1}^{N} \sum_{i=1}^{k} \pi_i N(x_j, \mu_i, \sigma_i^2))$

   C. both A and B

   D. None of these

2. (1 point) Given a HMM with N hidden states and M observation symbols, let $b_j(k)$ represent the probability of observing event j when in state k. Also, let π_j represent the probability of being in state j initially (π_j's represent the initial state probabilities). Which of the following statements is True?

   A. $\sum_{k=1}^{M} b_j(k) \geq 1$ and $\sum_{j=1}^{N} \pi_j \leq 1, \forall 1 \leq j \leq N.$

   B. $\sum_{j=1}^{N} b_j(k) = 1$ and $\sum_{j=1}^{N} \pi_j = 1, \forall 1 \leq k \leq M.$

   C. $\sum_{k=1}^{M} b_j(k) = 1$ and $\sum_{j=1}^{N} \pi_j = 1, \forall 1 \leq j \leq N.$

   D. $\sum_{j=1}^{N} b_j(k) \geq 1$ and $\sum_{j=1}^{N} \pi_j \leq 1, \forall 1 \leq k \leq M.$

3. (1 point) If $f(z) = tanh(z)$, then the derivative of $f(z)$ will be?

   A. $f(z)$

   B. $1 - f(z)$

   C. $1 - f^2(z)$

   D. $f^2(z)$

Options :

6406531515366. ✔ I have written answers on the answer sheets

6406531515367. ✖ Not applicable

Sub-Section Number :	3
Sub-Section Id :	64065365966
Question Shuffling Allowed :	No
Is Section Default? :	null

Question Number : 164 Question Id : 640653455740 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4

Question Label : Multiple Choice Question

4. (1 point) Recurrent Neural Networks(RNN) with basic RNN cells exhibit which of the following problems?

   A. Vanishing gradient problem

   B. Exploding gradient problem

   C. both (A) and (B)

   D. None of the above

5. (1 point) Which of the following ideas is used by LSTM-based RNN networks to capture long range dependencies?

   A. Pooling

   B. Convolution

   C. Gates

   D. None of the above

6. (1 point) State whether the following statement is true or false: A Transformer Encoder layer can compute the hidden representation of all time steps simultaneously.

   A. True

   B. False

7. (1 point) Which of the following is correct according to Bayes rule?

   A. $P(A|B) = \frac{P(A)}{P(B)}$

   B. $P(A|B) = \frac{P(A \cap B)}{P(A)}$

   C. $P(A|B) = \frac{P(A \cap B)}{P(B)}$

   D. $P(A|B) = \frac{P(A \cup B)}{P(B)}$

Options :

6406531515368. ✔ I have written answers on the answer sheets

6406531515369. ✖ Not applicable

Sub-Section Number : 4

Sub-Section Id : 64065365967

Question Shuffling Allowed : No

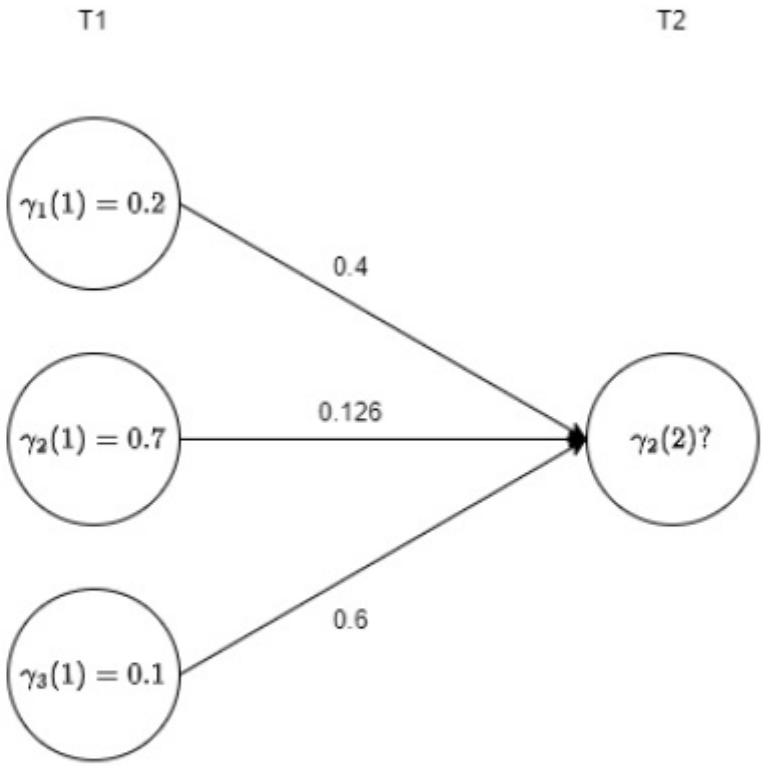
Is Section Default? : null

Question Number : 165 Question Id : 640653455741 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1

Question Label : Multiple Choice Question

8. (1 point) Consider the Fig, and determine $\gamma_2(2)$



A. 0.08
B. 0.0882
C. 0.006
D. 0.2282

Options :

6406531515370. ✔ I have written answers on the answer sheets

6406531515371. ✖ Not applicable

Sub-Section Number : 5

Sub-Section Id : 64065365968

Question Shuffling Allowed : No

Is Section Default? : null

Question Number : 166 Question Id : 640653455742 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 8

Question Label : Multiple Choice Question

1.2 Subjective Questions

1. (2 points) Draw the architecture diagram of the LSTM and explain each component briefly.
2. (3 points) Which of the following are valid activation functions you could use to train neural networks. Draw graphs representing the result of using activation function at the input, similar to graphs for activation functions shown in the class. Justify your answer.
   - $f(x) = min(0,\ x)$
   - $f(x) = 0.7x$
   - $f(x) = \begin{cases} 1 & if\ \ x > 0.5 \\ -1 & else \end{cases}$
3. (3 points) Explain the difference between local and global minima. Discuss the steps suggested in literature to avoid problem of getting stuck at a local minima.

Options :

6406531515372. ✔ I have written answers on the answer sheets

6406531515373. ✖ Not applicable

Sub-Section Number : 6

Sub-Section Id : 64065365969

Question Shuffling Allowed : No

Is Section Default? : null

Question Number : 167 Question Id : 640653455743 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 14

Question Label : Multiple Choice Question

4. (3 points) we discussed a technique ($Word2Vec$) to obtain word embeddings given a text corpus. We noted that these representations have interesting properties like syntactically similar words are 'closer' to each other compared to dissimilar words. Say we wish to extend this idea to obtain representations for words across 2 languages (English and French). Further we wish to have similar words across language to have representations that are 'close'. We are given a corpus of English and French text. We are also given a dictionary D of words that maps words in English to equivalent words in French. Write down the objective function for training a $Word2Vec$ model for the above task. A good language model assigns higher probability for "valid" sentences in a language and low probabilities for sentences which are incorrect (for instance - sentences which do not follow language grammar rules).

5. (3 points) Describe the RNN encoder-decoder block for machine translation and how the attention mechanism is embedded. List down the three popular techniques for calculating attention.

6. (3 points) We saw that for a given decoder timestep t, the attention weights over the S encoder states (timesteps) are constrained to sum up to 1 ($i.e., \sum_{t=1}^{S} \alpha_{it} = 1$). However, there is no constraint on how the attention weights behave across timesteps. This could result in a situation where the attention weights are always concentrated on the same set of encoder states across different decoder timesteps. This may be undesirable (for example, in the case of translation this might mean that some words in the source sentence never get attention). Suggest a way to tweak the loss function to explicitly prevent this problem?

7. (5 points) Draw the block diagram of the Transformer network. Explain all the components in detail.

Options :

6406531515374. ✔ I have written answers on the answer sheets

6406531515375. ✖ Not applicable

Sub-Section Number :	7
Sub-Section Id :	64065365970
Question Shuffling Allowed :	No
Is Section Default? :	null

Question Number : 168 Question Id : 640653455744 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 10

Question Label : Multiple Choice Question

8. (5 points) Consider a weather prediction based HMM problem with three hidden states: *Sunny, Cloudy and Rainy.* There are four different observable states that depend on the hidden states, namely: *Dry, Dryish, Damp, and Soggy.* The transition and observation probabilities are shown in fig. Given initial Cloudy weather condition on Day0. It was observed that the weather was *Dry* followed by *Damp* for the next two days. Compute the best possible weather forecast sequence which best explains these two observations (*hint: Use Viterbi Algorithm*).

Grass

Weather	Dry	Dryish	Damp	Soggy
Sunny	0.60	0.20	0.15	0.05
Cloudy	0.25	0.25	0.25	0.25
Rainy	0.05	0.10	0.35	0.50

Today

Yesterday	Sunny	Cloudy	Rainy
Sunny	0.5	0.375	0.125
Cloudy	0.25	0.125	0.625
Rainy	0.25	0.075	0.675

9. (5 points) We discussed Bidirectional Encoder Representations from Transformer (BERT) as a way to obtain embeddings, which can be helpful in many NLP tasks. List the mechanism used for training the BERT model and highlight differences with the *Word2Vec* training.

Options :

6406531515376. ✔ I have written answers on the answer sheets

6406531515377. ✖ Not applicable

Design Thinking

Section Id :	64065329452
Section Number :	8
Section type :	Online
Mandatory or Optional :	Mandatory